

Roll No.

TMA-401

B. TECH. (EEE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

MATHEMATICS—III

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) A bag contains four black and five red balls. If three balls from the bag are chosen at random, what is the chance that they are all black ?

(CO1)

- (b) An urn contains 10 white, 15 red and 25 black marbles. Determine the probability of drawing :

(CO1)

(i) Either a white or a red marble.

(ii) Either a white or a red or a black marble.

P. T. O.

- (c) If 12 persons are seated at a round table, what is the probability that two particular persons sit together ? If 6 boys and 6 girls sit in a row randomly, find the probability that all the 6 girls sit together. (CO1)
2. (a) Let X be a continuous random variable with probability density function : (CO2)

$$f(x) = \begin{cases} kx & ; 0 \leq x \leq 1 \\ k & ; 1 \leq x \leq 2 \\ -kx + 3k & ; 2 \leq x \leq 3 \\ 0 & ; \text{elsewhere} \end{cases}$$

- (i) Determine the value of k .
- (ii) Find mean and variance of X .
- (b) The random variable X has a Poisson distribution. If $P(X = 1) = 0.01487$, $P(X = 2) = 0.04461$, then find $P(X = 3)$. (CO2)
- (c) Three machines A, B, C produce respectively 60%, 30%, 10% of the total number of items of factory. The percentages of defective output of this machine are respectively 2%, 3%, 4%. An item is selected at random from the product and is found defective. Find the probability that item was produced by machine C. (CO2)
3. (a) If X and Y have joint p. d. f. : (CO3)

Y \ X	X		
	1	2	3
1	0.1	0.1	0.2
2	0.2	0.3	0.1

Find :

- (i) Marginal distribution of X and Y
- (ii) Conditional distribution of X given $Y = 1$
- (iii) Evaluate $P[x + y < 4]$.

- (b) Toss 3 coins. Let X denotes the number of heads on the first two and Y denotes the number of heads on the last two. (CO3)

- (i) Find the joint distribution of X and Y .
- (ii) Find $E[y/x = 1]$.
- (iii) Find the correlation coefficient $\rho(x, y)$.

- (c) The marks obtained by the candidates in Statistics in a certain examination are found to be normally distributed. If 12.5% of the candidates obtain 60 or more marks and 39% obtain less than 30 marks, find the mean number of marks obtained by the candidate. (CO3)

4. (a) Calculate the first four moments and define Kurtosis from the following data : (CO4)

x	f
0	5
1	10
2	15
3	20
4	25
5	20
6	15

P. T. O.

- (b) In Miss India Contest ten competitors are ranked by three judges in the given table. Use the correlation coefficient to determine which pair judge has the nearest approach to common perception in beauty : (CO4)

1st Judge	2nd Judge	3rd Judge
5	1	6
3	6	4
10	5	9
7	10	8
2	3	1
1	2	2
4	4	3
10	9	10
4	7	5
6	8	7

- (c) The following figure relate to the number of limit producer per shift by two workers A and B a number of day : (CO4)

A	B
19	26
22	37
24	40
27	35

24	30
18	30
20	40
19	26
25	30
	35
	45

Can it be inferred that A is more stable worker compared to B ? Answer using F-test at 5% level of significance.

{From Table $F_{10, 8 (0.05)} = 3.35$ }.

5. (a) The following data relate to the intake of proteins and gain in body weight of 10 albino rats : (CO5)

Serial No. of Rats	Protein intake (gm)/day	Gain in weight (gm)
1	10	12
2	11	15
3	12	16
4	14	22
5	16	24
6	13	18

7	15	23
8	12	14
9	13	20
10	17	25

Find the value of correlation coefficient and analyse the result.

- (b) A soap manufacturing company was distributing a particular type of brand through a large number of retail shops. Before a heavy advertising campaign, the mean sales per week per shop was 140 dozen. After the campaign, a sample of 26 shops was taken and the mean sales was found to be 147 dozen with standard deviation 16. Do we consider that the advertisement effective for increasing sales ? (The critical value of $t_{2,5} (0.05) = 1.71$). (CO5)

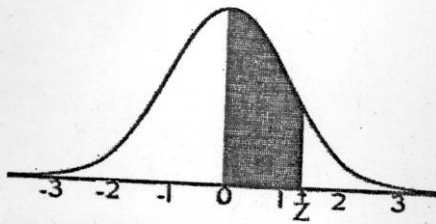
- (c) A garage has recorded the following faults in cars of four makes over a period of time : (CO5)

Make	Nature of faults		
	Mech.	Electrical	Chasis
A	32	26	12
B	60	54	22
C	52	38	15
D	28	25	12

Can you say the type of fault is independent of the make of the car ?

(Given $\chi^2 = 12.592$, for 6 degrees of freedom).

This table is required for Q. 3 (c).



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

Roll No.

TEE-401

**B. TECH. (EE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

ELECTRICAL MACHINES-II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the role of harmonics on induced emf of synchronous generator.
(CO1, CO3)
- (b) Derive the relation of induced emf for synchronous generator including the effects of pitch factor and distribution factor. (CO1, CO3)
- (c) A 3 phases, 16 pole synchronous generator has star connected winding with 144 slots and 10 conductors per slot. The flux per pole is 0.04 Wb (sinusoidally distributed) and the speed is 375 rpm. Find the frequency and phase and line emf. Total turn may be assumed to be series connected. (CO1, CO3)
2. (a) Discuss torque slip characteristics of Induction motor. (CO2, CO6)

P. T. O.

- (b) A 460-V, 60-Hz, 80-hp, three-phase induction motor is drawing 60 A at 0.85 PF lagging. The stator copper losses are 2 kW, and the rotor copper losses are 700 W. The friction and windage losses are 600 W, the core losses are 1800 W, and the stray losses are negligible. Find the following quantities : (CO2, CO6)
- (i) The air-gap power P_{AG}
 - (ii) The power converted P_{conv}
 - (iii) The output power P_{out}
 - (iv) The efficiency of the motor
- (c) Draw equivalent circuit of an induction motor and hence find the expression for gross mechanical power developed. (CO2, CO6)
3. (a) (i) Why is it impossible for an induction motor to operate at synchronous speed ?
- (ii) How does a resistive starter circuit for an induction motor work ? (CO4, CO6)
- (b) Discuss three phase rotating magnetic field with a suitable diagram. (CO4, CO6)
- (c) Discuss V/f control for an induction motor. (CO4, CO6)
4. (a) The full load current of a 3.3 kV star connected synchronous motor is 160 A at 0.8 pf lagging. The resistance and reactance of the motor are 0.8Ω and 5.5Ω per phase respectively. Calculate the excitation emf, torque angle, efficiency and shaft output of the motor. Assume mechanical stray losses to be 30 kW. (CO4, CO5)
- (b) Discuss OCC and SCC of synchronous generator. (CO4, CO5)

(3)

- (c) Derive the expression for power angle characteristics and discuss it in details. (CO4, CO5)
5. (a) A 3300 V, delta connected motor has synchronous resistance per phase (delta) of 18Ω . It operates at leading power factor of 0.707 when drawing 800 kW from mains. Calculate excitation emf. (CO6, CO5)
- (b) Discuss the principle of working of single phase reluctance motor with a suitable diagram. (CO6, CO5)
- (c) Explain the double-revolving-field theory of single-phase Induction Motors. (CO6, CO5)

Roll No.

TEE-402

B. TECH. (EEE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022
INTRODUCTION TO ELECTRICAL ENERGY SOURCES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How are boilers classified ? Why is ash handling plant necessary in a thermal plant ? (CO1)
- (b) A steam power station of 100 MW capacity uses coal of calorific value 6400 kcal/kg. The thermal efficiency is 30% and electrical generation efficiency is 92%. Find coal required per hour when plant is working at full load. (CO1)
- (c) Draw and explain water steam flow diagram of a thermal plant. (CO1)

P. T. O.

2. (a) Distinguish among fast and thermal reactors, fissile and fertile material, fusion and fission. (CO2)
- (b) Find the power produced by fissioning 5 grams of U^{235} per day. (CO2)
- (c) Give the main features of (i) Pressurised water reactor and (ii) Boiling water reactor. (CO2)
3. (a) How can hydro-plants be classified according to (i) water flow regulation, (ii) head and (iii) load ? (CO3)
- (b) The total water inflows for different months for a river are as under : (CO3)

January	$1000 \times 10^6 \text{ m}^3$
February	$800 \times 10^6 \text{ m}^3$
March	$550 \times 10^6 \text{ m}^3$
April	$300 \times 10^6 \text{ m}^3$
May	$300 \times 10^6 \text{ m}^3$
June	$1500 \times 10^6 \text{ m}^3$
July	$3000 \times 10^6 \text{ m}^3$
August	$3000 \times 10^6 \text{ m}^3$
September	$2500 \times 10^6 \text{ m}^3$
October	$1500 \times 10^6 \text{ m}^3$
November	$1450 \times 10^6 \text{ m}^3$
December	$1600 \times 10^6 \text{ m}^3$

- (i) Plot hydrograph and flow duration curve. Find out average monthly flow.

- (ii) Find power available at mean flow if head available is 100 m and efficiency of generation is 80%.
- (c) Differentiate between the operation of a Pelton turbine and a Francis turbine. (CO3)
4. (a) Which fuels are used in diesel electric plants ? Discuss the main parts of a diesel power plant. (CO4, CO5)
- (b) Previously diesel and gas turbine plants were used only for peaking duty. Now they are also used as base load plants. Why ? (CO4, CO5)
- (c) Discuss the working of Gas Turbine Plant with the help of a suitable diagram. (CO4, CO5)
5. (a) What is tariff ? Explain the various types of tariffs. (CO5, CO6)
- (b) Incremental fuel cost in rupees per megawatt hour for two units in a plant are given by the following two equations :

$$0.1 P_1 + 20 \text{ and } 0.12 P_2 + 16$$

The minimum and maximum load on each unit is to be 20 MW and 125 MW. Determine incremental fuel cost and allocation of load between units for minimum cost when load is 100 MW. (CO5, CO6)

- (c) A residence has the following connected loads : 6 lamps to 100 watt each, 6 lamps of 40 watt each, 5 fans of 60 watt each, 2 power plug points 1000 watt each, 4 light plug points 100 W each and one geyser 3000 watt. The electricity use on a typical summer day can be described as under.

Lamps – One 100-watt lamp from 5 a.m. to 6 a.m., 3 lamps each of 100 watt and 3 lamps of 40 watt from 7 p.m. to 11 p.m.

P. T. O.

Fans—Two fans used simultaneously from 8 a.m. to 11 p.m.

Power plug load—Nil

Light Plug load—One point used from 6 a.m. to 8 am, 12 Noon to 2 p.m. and 7 p.m. to 10 p.m.

Geyser—9 a.m. to 10 a. m.

Draw chronological load curve for this residence for summer.

Find : (i) maximum demand, (ii) demand factor, (iii) average load and (iv) monthly consumption of electricity. (CO5, CO6)

Roll No.

TEE-403

**B. TECH. (EE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

SIGNALS AND SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

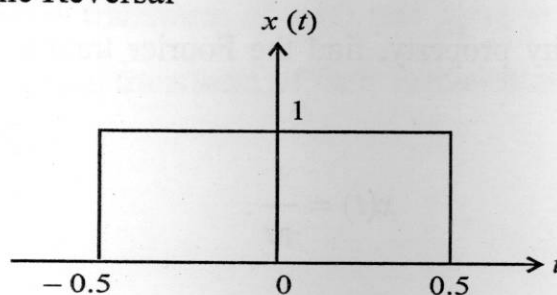
1. (a) Discuss the following transformation properties considering a rectangular signal of width and height of 1 unit as shown in figure :

(CO1)

(i) Time Shifting

(ii) Time Scaling

(iii) Time Reversal



P. T. O.

(b) Determine the fundamental period of the following signals : (CO1)

(i) $x(t) = 5 \sin 100 \pi t + 10 \sin 50 \pi t$

(ii) $x(t) = \text{event part of signal } [\cos(4\pi t) u(t)]$

(iii) $x(t) = \text{event part of signal } [\sin(4\pi t) u(t)]$

(c) Prove that following property of continuous-time unit impulse function : (CO1)

(i) Shifting Property

(ii) Scaling Property

(iii) Differentiation Property

2. (a) If $x(t) * h(t) = y(t)$, then show that : (CO2)

$$x(t) * h(t - t_0) = x(t - t_0) * h(t) = y(t - t_0)$$

(b) If $x(t) * h(t) = y(t)$ then show that $x(at) * h(at) = \frac{1}{|a|} y(at)$. (CO2)

(c) Find the convolution of two continuous-time signals $f(t) = e^{-t^2}$ and $g(t) = 3t^2$ for all t . (CO2)

3. (a) Derive exponential form of Fourier series from its trigonometric form and line magnitude and phase spectrum. (CO3)

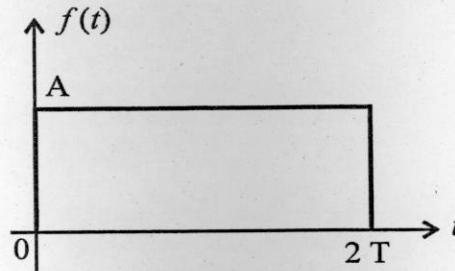
(b) (i) If Fourier transform of $x(t)$ is $g(\omega)$ then show that Fourier transform of $g(t)$ will be $2\pi x(-\omega)$.

(ii) Using duality property, find the Fourier transform of signal given below : (CO3)

$$x(t) = \frac{1}{\pi t}$$

(3)

- (c) Derive the Fourier transform of the pulse as shown below : (CO3)



4. (a) Find the Laplace transform of the following signals : (CO4)

- (i) t^n
- (ii) $\sin \omega t$
- (iii) $\sin hbt$

- (b) Find Inverse of Laplace transform of the following functions : (CO4)

(i) $F(s) = \frac{se^{-2s}}{s^2 + 2s + 5}$

(ii) $F(s) = \frac{s + 4}{2s^2 + 5s + 3}$

- (c) State and prove Initial and Final value theorem. (CO4)

5. (a) Discuss sampling theorem. (CO5, CO6)

- (b) Determine the Nyquist rate of the following signals : (CO5, CO6)

- (i) $x(t) = \sin 200 \pi t$
- (ii) $x(t) = \sin^2 200 \pi t$
- (iii) $x(t) = \cos(150 \pi t) \times \sin(100 \pi t)$

- (c) If Laplace transform of $x_1(t)$ and $x_2(t)$ are $X_1(s)$ and $X_2(s)$ then show that Laplace transform of their convolution will be the product of $X_1(s)$ and $X_2(s)$. (CO5, CO6)

Roll No.

TEE-404

**B. TECH. (EE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

DIGITAL ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) State and prove any *five* Boolean identities. (CO1)
(b) Write counting in BCD code and in hexadecimal from 0 to 20. (CO1)
(c) State and prove Consensus theorem. (CO1)
2. (a) Simplify Boolean function
 $Y = \text{minterms } (1, 3, 4, 5, 7, 8, 9, 11, 12, 14, 15).$ (CO2)
(b) Prove that NOR gate is a universal gate. (CO2)
(c) Design a 4 bit binary adder-subtractor combinational circuit. (CO2)
3. (a) Design a 16-to-1 multiplexer using five 4-to-1 multiplexers. (CO3)

P. T. O.

(2)

- (b) Explain half subtractor and full subtractor with the help of truth table and circuit design. (CO3)
- (c) Design a 3-bit magnitude comparator and explain its logic. (CO3)
- 4. (a) Derive characteristic equation for T flip flop and for D flip flop. (CO4)
- (b) Design a JK flip flop using NAND gates. Mention its action table, characteristic table and excitation table. (CO4)
- (c) Differentiate between combinational and sequential circuits. (CO4)
- 5. (a) Design a mod 6 synchronous up counter using T flip flop. (CO5, CO6)
- (b) Design a 4 bit ring counter using D flip flop. (CO5, CO6)
- (c) Explain various steps of designing a synchronous counter. (CO5, CO6)